

Assignment 2

- A) We write up the utility functions for the mixed strategies and check if they intersect in the pure strategies

$$V_1(p, q) = (2 - 2p) \cdot q + p \cdot (1 - q) =$$

Player 1 sets p to maximize V for every q

We differentiate with respect to p for player 1

$$\frac{\partial V_1}{\partial p} = -2q + 1 - q = 1 - 3q$$

If larger than 0 we set price as high as possible and vice versa

If equal to 0 player 1 is indifferent to the price level

This implies a best response function

$$B_1(q) = \begin{cases} 0 & \text{for } q < \frac{1}{3} \\ [0,1] & \text{for } q = \frac{1}{3} \\ 1 & \text{for } q > \frac{1}{3} \end{cases}$$

We do the same for player 2

$$V_2(p, q) = p \cdot q + (1 - p)(1 - p) = 1 - q - p + 2 \cdot q \cdot p$$

$$\frac{\partial V_2}{\partial q} = -1 + 2p$$

If larger than 0 player 2 chooses to showcase if lower chooses not to showcase

In mixed strategies this would be a $q=1$ or $q=0$

Again this gives the best response function

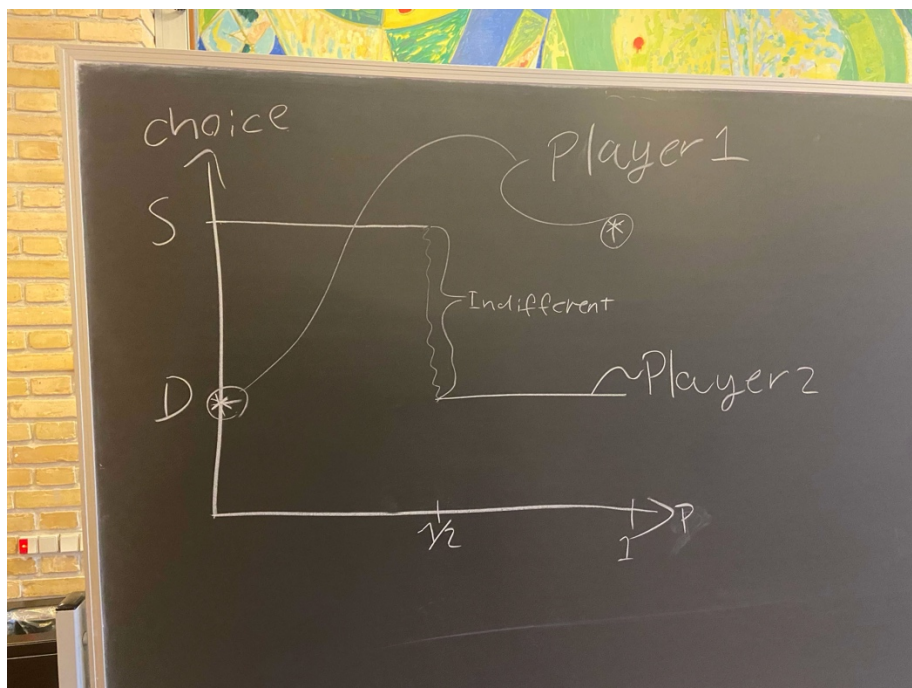
$$B_2(p) = \begin{cases} S & \text{for } p < 1/2 \\ S \text{ or } D & \text{for } p = \frac{1}{2} \\ D & \text{for } p > \frac{1}{2} \end{cases}$$

Translated to probabilities

$$B_2(p) = \begin{cases} 1 & \text{for } p < 1/2 \\ [0,1] & \text{for } p = \frac{1}{2} \\ 0 & \text{for } p > \frac{1}{2} \end{cases}$$

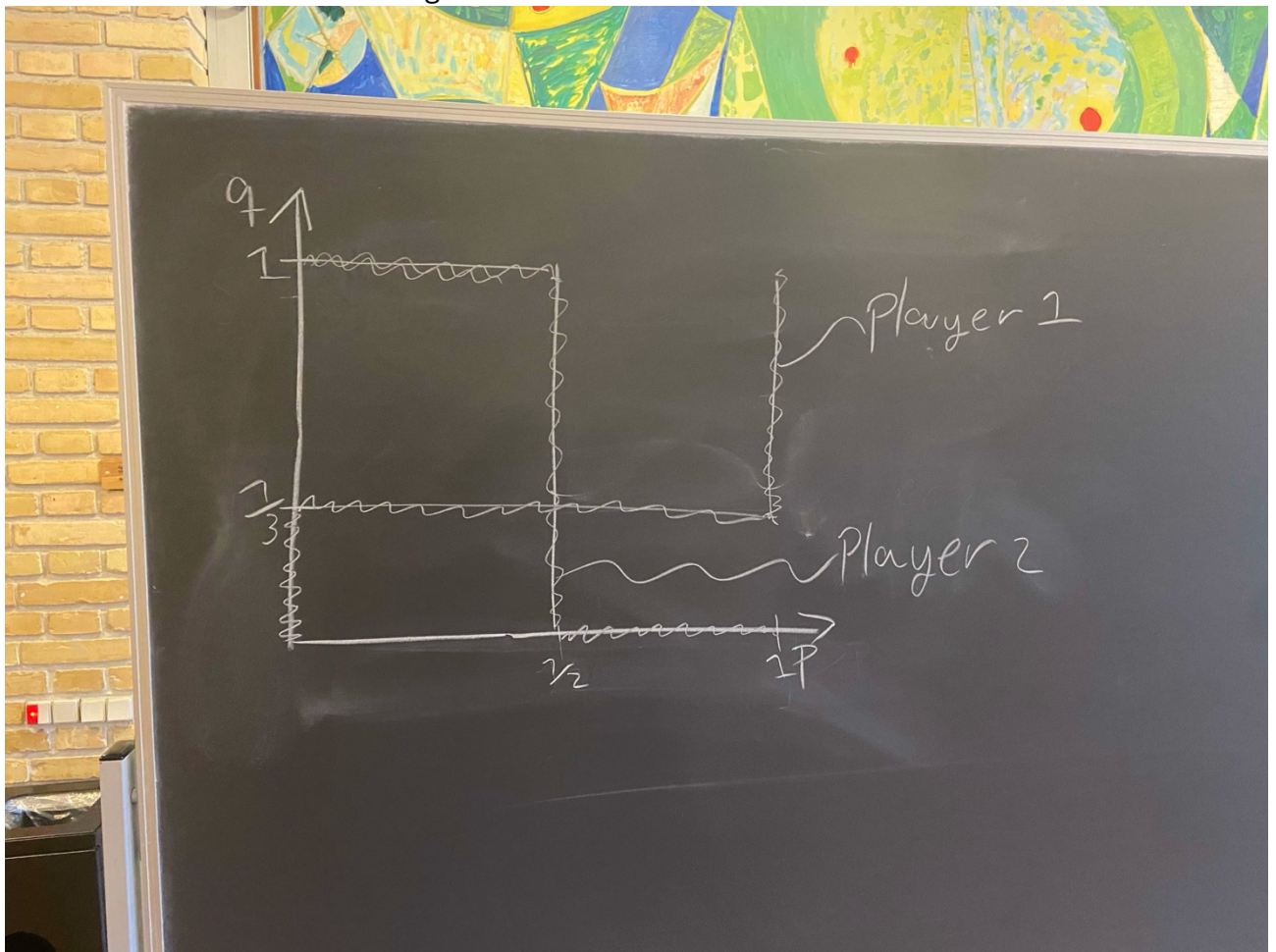
We draw these 2 different best response diagrams

First with player 2 not being able to mix



Since player 2 is not able to mix player 1's response function is not defined between the 2 choices and thus the best response functions never intersect and thus we have no Nash EQ

Second when he can mix strategies



When looking at the best response diagram we identify NE by intersections of the best responses

We have an intersection at $(p, q) = (\frac{1}{2}, \frac{1}{3})$

Where we have a mixed nash eq

$$MSEQ = \left\{ \left(p = \frac{1}{2} \right); \left(q = \frac{1}{3} \right) \right\}$$

We calculate each players expected payoff in Nash

$$V_1 \left(\frac{1}{2}, \frac{1}{3} \right) = \left(2 - \frac{2}{2} \right) \cdot \frac{1}{3} + \frac{1}{2} \cdot \left(1 - \frac{1}{3} \right) \approx 0,6666667$$

$$V_2 \left(\frac{1}{2}, \frac{1}{3} \right) = 1 - \frac{1}{3} - \frac{1}{2} + 2 \cdot \frac{1}{3} \cdot \frac{1}{2} = 0,5$$